2019 IEEE International Conference on Service-Oriented System Engineering (SOSE)

Multi-Objective Path Planning for UAV in the Urban Environment Based on CDNSGA-II

1st Qian Ren
School of Computer Science
Northwestern Polytechnical University
Xi'an, China
renqian@mail.nwpu.edu.cn

2nd Yuan Yao
School of Computer Science
Northwestern Polytechnical University
Xi'an, China
yaoyuan@nwpu.edu.en

3rd Gang Yang
School of Computer Science
Northwestern Polytechnical University
Xi'an, China
yeungg@nwpu.edu.cn

4th Xingshe Zhou
School of Computer Science
Northwestern Polytechnical University
Xi'an, China
zhouxs@nwpu.edu.cn

***Abstract*—This paper is concerned with path planning for unmanned aerial vehicles (UAVs) in urban environment. Distance is the fundamental goal of path planning. Besides, safety is indispensable for the riskless UAV flight. In this paper, we propose a multi-objective path planning (MOPP) approach based on Non-dominated Sorting Genetic Algorithm II (NSGA-II) to find an optimal collision-free path for UAV, considered both distance and safety. To this aim, firstly, we represent urban environment by octree, which is an efficient and hierarchical data structure for spatial subdivision in 3D. Meanwhile, safety index map (SIM) are developed to capture obstacles in the geography map. Then, a CDNSAG-II methods proposed to deal with MOPP problem. The experimental results demonstrate that our approach is able to find an optimal path efficiently under urban environment based on octrees.**

***Index Terms*—UAV, multi-objective path planning, NSGA-II, octrees, urban environment**

I. Introduction

Recently, the usage of UAVs has gained extensive attention around the world [1]. UAV playes an important role in a variety of aspects such as cargo delivery and aerial photo. To benefit these usages of a UAV, it should be safely and timely reach the target area, namely, path planning problem.

Path planning is the process of searching a collision-free pathway between the start point and goal point. So far, three types of path planning methods have been proposed, namely, the geometry method [2], the velocity space method [3], and the graph search method [4].

However, these path planning approaches cannot meet the goal, flying timely and safely, so that they cannot be directly exploited to perform the UAV path planning under urban environment. A popular approach is based on octrees for the representation of three-dimensional environments. And, it is efficient to update and search space while keeping the memory consumption at a minimum.

In this paper, firstly, we use the representation of environments based on octrees. On the basis of it, we introduce a safety index map (SIM) which shows the location of obstacles and the safety index of every points in map. Then, we proposed Crowd Distance based NSGA-II (CDNSGA-II) method to cope with the MOPP problem. We leverage A* search algorithm to initialize group. Meanwhile, we utilize non-dominated sorting to select group.

The rest of this paper is as follows. In section 2, we introduce the related works. Section 3 models the proposed problem and section 4 describes the CDNSGA-II and presents its detailed design and implementation. We show our experimental results in section 5 and conclude this paper in section 6.

II. Related Work

The multi-objective path planning playes an important role in QoS routing, motion design and navigation. It is well known that the path planning with multi-objective is an NP-complete problem [5].

In general, there are two main methods to deal with the Multi-Objective Optimization Problem (MOOP). The one method converts a multi-objective problem into a single-objective problem by using a weighted-sum function. This method is simple, but weights for different objectives are determined before, which is difficult. And, the bias will be imposed during the whole optimization process. Thus, the solution of the first method is often non-optimized.

Another method first gets a set of Pareto optimal solutions. Then, the most suitable one is selected from the Pareto optimal set, which is easy by the first method. And, it is not different to design the weight vector by normalizing all objectives into the same order of magnitude. The method can be classified into two types: exact method and heuristic method.

Some exact methods are based on the Dijkstra's algorithm to get exact Pareto optimal solutions for path planning [6]. However, the computational complexity is extremely high. The Genetic Algorithm (GA) based on meta-heuristics has emerged as an effective method to solve the MOOP problem. Many multiple objective GA are proposed such as the Niched Pareto Genetic Algorithm (NPGA) [7], the Multi-Objective Genetic

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DOI 10.1109/SOSE.2019.00059

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Algorithms (MOGA) [8], the Strength Pareto Evolutionary Algorithm 2 (SPEA2) [9] and the Non-dominated Sorting Genetic Algorithm II(NSGA-II) [10]. Among all above algorithms, NSGA-II presents superior performance on not only benchmark problems, but also real-world applications [11]. Therefore, many work adopt NSGA-II in the multi-objective path planning problem.

Meanwhile, there are various works to address the MOPP problem for UAVs. For example, Hernndez-Hernndez et al. proposed a multi-objective A* algorithm to find a UAV path, where travel time, path angle, fuel consumption and area deviation are considered [12]. Tao et al. presented a modified GA to solve the MOPP problem. [13]. In addition, Niendorf et al. proposed a method to cope with the UAV mission planning in hostile environments, where a tradeoff was made between short mission execution time and low risk [14].

Although these MOPP algorithms have been developed to find an suitable path for a UAV flying in a complex environment, none of them has considered the efficient representation of environments.

Three-dimensional models of the environment are a key prerequisite for many robotic systems. Consequently, they have been the subject of research for more than two decades. As a data structure, octrees are applied in a variety of applications, most notably in the area of computer graphics for efficient rendering and in the field of photogrammetry to store and address large point clouds.

III. Problem Definition

Since we need to deal with an MOPP problem in this paper, we first describe a general MOPP problem by the theory of graph. The path planning is defined as finding an optimal path p between the source and destination points in a graph $G(V, E)$ considering two objectives: distance and safety.

V is a set of points in space, denoted by $\{v_1, v_2, ..., v_N\}$. And, E is a set of edges connecting two points in V, denoted by $\{e_1, e_2, ..., e_M\}$. A path p is a sequence of points $\{v_1, v_2, ...v_h\}$ from the source point v_1 to the destination point v_h. $c_{i,j}$ is the non-negative cost of the jth objective assigned for the ith edge ($i \in [1, M]$ and $j \in [1, K]$, K is the number of objectives). Specifically, K equals to 2 in this paper, $c_{i,1}$ and $c_{i,2}$ are costs of the ith edge for distance and safety index respectively. Let $f_j(p)$ be the total cost of all edges in path p for the jth ($j \in [1, K]$) objective. Then, $f(p)$ is the objective vector for path p: $f(p) = \langle f_1(p), f_2(p), ...f_K(p)\rangle$.

As defined, a feasible path is a sequence of points from the source point to the destination point. Obviously, there are many feasible paths for a given source and destination pair in a graph. The MOPP problem is to find an optimal one. Generally, we make a trade-off among all objectives and try to obtain Pareto optimal solutions.

Definition 1. Pareto Optimal Solution: For a path p and a path q, when they satisfies the following conditions:

$$\begin{aligned} &f_j(p) \leq f_j(q), \forall j \in [1, K] \\ &\exists i \in [1, K], f_i(p) < f_i(q). \end{aligned} \tag{1}$$

We say that p dominates q, denoted by $p \prec q$. If there is no path dominates p, p is called a non-dominated solution, i.e. a Pareto optimal solution. The other dominated solutions are called feasible solutions.

Definition 2. Pareto Optimal Set (Pareto Front): A Pareto optimal set (denoted by $\mathbb{P}$) belonging to a feasible solution set (denoted by $\mathbb{F}$), and satisfies the following conditions:

$$\forall p \in \mathbb{P}, \mathbb{P} \subseteq \mathbb{F}, \nexists x \in \mathbb{F}, x \prec p. \tag{2}$$

The representation of the Pareto optimal set in the objective space is Pareto front as shown in Fig. 1.

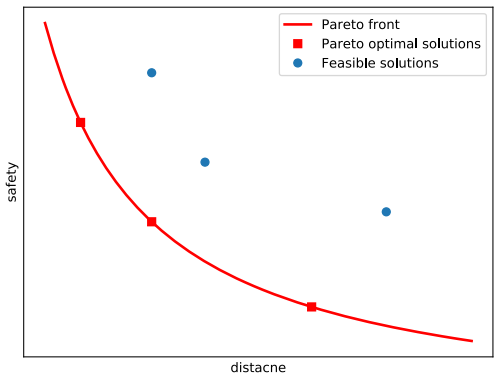


Fig. 1: The Pareto Front

Therefore, the MOPP problem is to find one from the Pareto optimal set $\mathbb{P}$ such that the total cost C is minimized.

$$C = \min \sum_{j=1, p \in \mathbb{P}}^{K} \alpha_j f_j(p), \sum_{j=1}^{K} \alpha_j = 1, \tag{3}$$

where $\alpha_j \in [0, 1]$ is the weight of the jth objective.

IV. Proposed CDNSGA-II for MOPP of UAV

In this section, we introduce the proposed CDNSGA-II approach to multi-objective path planning of UAV.

A. Environment Representation

1) Octree-represented Environment: An octree is a hierarchical data structure for spatial subdivision in 3D. Each node in an octree represents the space contained in a cubic volume, usually called a voxel. This volume is recursively subdivided into eight sub-volumes until a given minimum voxel size is reached. Generally, there are three types of voxel, black voxel, white voxel and grey voxel, meaning occupied, free and mixed respectively.

In a straight-forward octree implementation, each node in the tree stores the coordinate of its center location, its voxel size, and points to its children, which can be directly achieved by using eight pointers per node. However, this method leads to a significant memory overhead.

We overcome that by using one child pointer per node that points to an array of eight pointers as shown in Fig. 2. This

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array is only allocated if the node indeed has children and is not allocated for leaf nodes. Thus, any leaf node in the octree only stores the data itself and one (null) pointer.

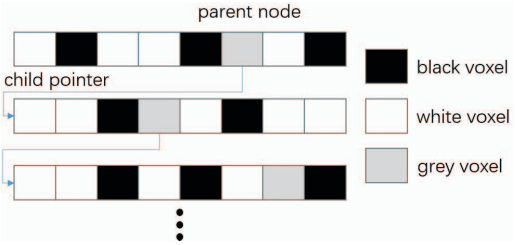


Fig. 2: One implementation of octree

2) Safety Index Map: Generally, the safety index of a point is associated with the distance between itself and obstacles. We could cope with the problem by utilizing multivariate normal random variable, and safety index could be calculated by probability density function.

$$f(x) = \frac{1}{\sqrt{(2\pi)^k \det \Sigma}} \exp\left[-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right], \tag{4}$$

where μ is the mean, k is the dimension of the space where x takes values and μ is the location of an obstacle's point. Σ is the covariance matrix represented by

$$\Sigma = E\left(\frac{r}{2}\right)^2, \tag{5}$$

where E is the identity matrix and r is the safety distance.

To obtain a normalized safety index of a point, we calculated the ratio of its probability density to maximum. Thus, we can obtain the SIM of a point as shown in Fig. 3.

$$s(x) = \frac{f(x)}{f(\mu)} \tag{6}$$

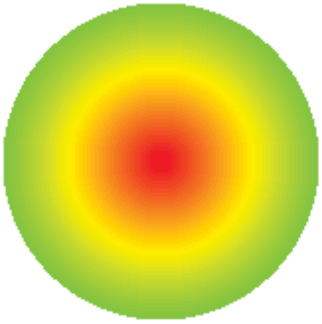


Fig. 3: One point SIM

The safety index of a point in the map is associated with all obstacles in the map.

$$s(x) = \max_{\mu \in o, o \in O} \frac{f(x)}{f(\mu)}, \tag{7}$$

where O is the set of obstacles, o the obstacle, and μ is a point constituting an obstacle. Fig. 4 shows the SIM of one obstacle.

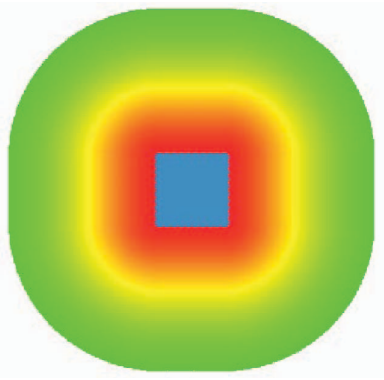


Fig. 4: One obstacle SIM

B. CDNSGA-II Algorithm

1) Initialization: In general, GA needs an initial population.we leverage the A* search algorithm with using a weighted-sum function to initialize, which it is more suitable than random generation.

The A* algorithm is known to be effective at finding the shortest path to a goal while avoiding obstacles. A* utilizes a cost heuristic function, which can take the form

$$f(\overrightarrow{p}) = g(\overrightarrow{p}) + h(\overrightarrow{p}), \tag{8}$$

Where $f(\overrightarrow{p})$ is the cost heuristic function, $g(\overrightarrow{p})$ the path cost from the start point to the current point, and $h(\overrightarrow{p})$ is the admissible heuristic estimation of the path cost from the current point to the target point.

Using a weighted-sum function, we calculate $g(\overrightarrow{p})$ and $h(\overrightarrow{p})$ by comprehensive cost instead of only distance cost. According to the size of initial group, different α could be selected, that distributed uniformly in domain of definition.

$$c_i = \alpha c_{i,1} + (1-\alpha) c_{i,2}, \alpha \in [0,1]. \tag{9}$$

2) Solution Evaluation and Selection: We use the non-dominated sorting algorithm to evaluate the quality of a given solution. As shown in Fig. 5, the Pareto optimal front is composed of a set of non-dominated solutions $\mathbb{P}_1$. We assign a front rank $\mathcal{F}(p) = 1$ for any path $p \in \mathbb{P}_1$. Then, we identify non-dominated paths in the rest solutions excluding paths $p \in \mathbb{P}_1$. These identified new non-dominated solutions are assigned with the front rank $\mathcal{F}(p) = 2$. The sorting process is continued until all solutions are classified into a certain front.

We also adopt the concept of crowd distance to compare two solutions in the same front. The NSGA-II only calculates the crowd distance considering solutions in the same non-dominated front. This cannot truly reveal the density of a point. For example, from Fig. 5. we notice that point A is more

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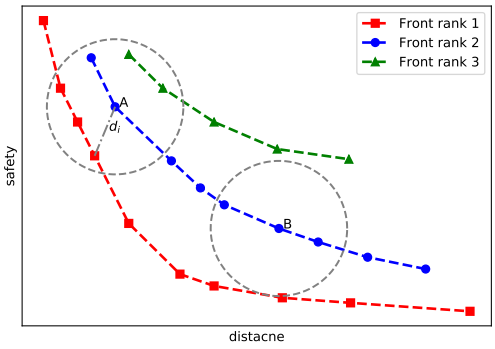


Fig. 5: Crowd Distance based Solution Selection

crowded than point B. But the NSAG-II will select point A because it is less crowded than B in the non-dominated front rank 2. To overcome this counter-intuitive solution selection, we redefine this distance as the average distance within k-nearest solutions.

$$\mathcal{D} = \frac{1}{k}\sum_{i=1}^{k} d_i. \tag{10}$$

Then, we can compare solutions according to these two attributes: non-dominated front rank $\mathcal{F}$ and crowd distance $\mathcal{D}$. We define a path p is superior to a path q if

$$\begin{aligned} &\mathcal{F}(p) < \mathcal{F}(q)\ OR \\ &\mathcal{D}(p) > \mathcal{D}(q)\ when\ \mathcal{F}(p) = \mathcal{F}(q). \end{aligned} \tag{11}$$

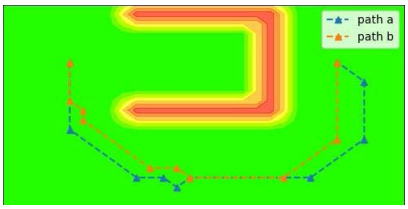
3) Crossover and Mutation: For the path p_i and p_j, ($p_i = \{v_{i,1}, ..., v_{i,k}, ..., v_{i,n}\}, p_j = \{v_{j,1}, ..., v_{j,l}, ..., v_{j,m}\}$) .If $v_{i,k}$ is equal to $v_{j,l}$, the crossover results will be p_i' and p_j' denoted by $\{v_{i,1}, ..., v_{j,l}, ..., v_{j,m}\}$ and $\{v_{j,1}, ..., v_{i,k}, ..., v_{i,n}\}$ respectively. When there are not only one equal points, we choose a point randomly.

There are two methods of mutation: change and remove. For the path p_i, we randomly choose a point $v_{i,k}$, and change it slightly or remove it with probability. An example is shown in Fig. 6. Fig 6a shows the original paths in the initial solutions. Fig 6b gives two new paths generated by crossover operation if the original paths have some interactions. Fig 6c gives the new path generated by mutation operation.

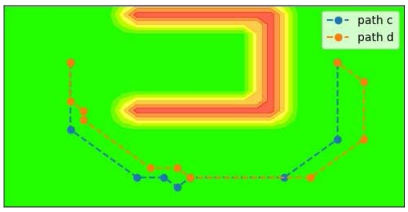
Summing up, the main step of CDNSGA-II can be described as the following.

V. Experiment Results

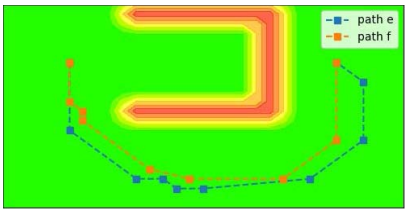
In this section, we carry out experiments to evaluate the performance of the proposed approach. We compare the NSGA-II [10] and the proposed CDNSGA-II in terms of solution quality. The exact Pareto optimal solutions are obtained by the Extended Martins' Label Setting (EMLS) [6] algorithm.



(a) Original paths



(b) New paths generated by crossover



(c) New paths generated by mutation

Fig. 6: An example of crossover and mutaion

A. Experiment Setup

We conduct experiments on an Intel core i5 machine of 8 GB RAM. The proposed algorithm is tested on the map of Beilin District in Xi'an, which has 5001 buildings (obstacles). The octree map is shown in the Fig. 7.

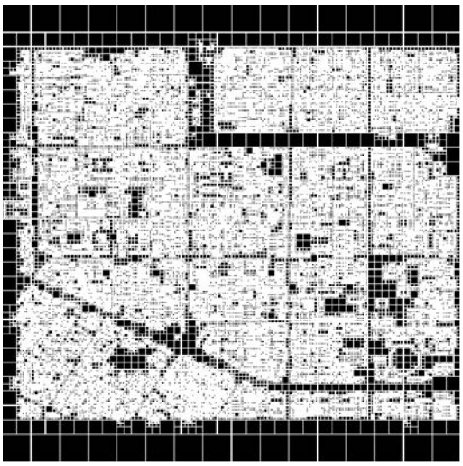


Fig. 7: The octree representation of Beilin District in Xi'an

The safety index is established by computing the distance to neighboring obstacles. For each unit of octree map, it has a value ranging from 0 to 1 that indicates the safe level of the unit (0 is very safe and 1 is very dangerous). We apply all

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Algorithm 1 CDNSGA-II

Input: p_s:Start Point, p_g:Goal Point, SIM:Safety Index Map.
Output: P_{op}: Optimal Path

$POP[N_s]$ = initialize by A* search algorithm;
while POP is not converged **do**
  for $i = 1 \rightarrow N_s$ **do** ▷ crossover(POP);
    for $j = i + 1 \rightarrow N_s$ **do**
      points[N_p] = intersection($POP[i]$,$POP[j]$)
      p = points[randint(N_p)]
      k_i, k_j = index($POP[i]$,p), index($POP[j]$,p)
      P_i = concatenate($POP[i][:k_i]$,$POP[j][k_j:]$)
      P_j = concatenate($POP[j][:k_j]$,$POP[i][k_i:]$)
      POP append P_i, P_j
    end for
  end for
  for $i = 1 \rightarrow N_s$ **do** ▷ mutation(POP);
    p = randget($POP[i]$)
    if $random() > rate$ **then**
      P_k = change($POP[i]$, p+random())
    else
      P_k = remove($POP[i]$, p)
    end if
    POP append P_k
  end for
  non-dominated sort POP ▷ selection(POP);
  select the best N_s POP
end while
P_{op} = getOptimalSolution(POP); **return** P_{op};

approaches to 3 areas from small to large size ($AREA_1$:415 obstacles, $AREA_2$:1530 obstacles, $AREA_3$:5001 obstacles). The SIM and the area partition are shown in Fig. 8.



Fig. 8: The SIM and area partition of Beilin District in Xi'an

To make the comparison as fair as possible, all algorithms adopt the same initial population size (N_s=100) and exactly the same initial paths. They will be stopped until either the loop reaches the maximum iteration number (I_{MAX}=200) or only one non-dominated front rank exists in the population and the $\Delta\mathcal{D}$ is smaller than a given predefined threshold (set to be 10 in experiments). The crossover rate and mutation rate are 0.9 and 0.2 respectively. To eliminate the impact of different magnitudes on metric measurements, all solutions of each objective are normalized to the range of $[0, 1]$ according to the min-max normalization.

$$\Delta\mathcal{D} = \overline{\mathcal{D}}\left(P_{t+1}\right) - \overline{\mathcal{D}}\left(P_t\right), \tag{12}$$

where $\overline{\mathcal{D}}(P)$ is the average crowd distance of all individuals in the population P.

In this paper, we compare the solution quality among A*, NSGA-II and the proposed CDNSGA-II.

Solution Quality: The optimal solution ratio (denoted by δ) is adopted to evaluate the accuracy of the proposed method. It defined as the proportion of exact optimal solutions to the total number of obtained solutions.

$$\delta = N_{exact}/N_{total}. \tag{13}$$

To evaluate the solution quality in a fine-grained manner, we define the average error (denoted by ρ) to measure the extent of convergence to exact Pareto optimal solutions. It is the average distance of σ_i, which is the minimum Euclidean distance from each obtained solution to the real Pareto optimal solution as shown in Fig. 9.

$$\rho = \frac{1}{N_s}\sum_{i=1}^{N_s}\sigma_i, \tag{14}$$

where N_s is the size of the initial population.

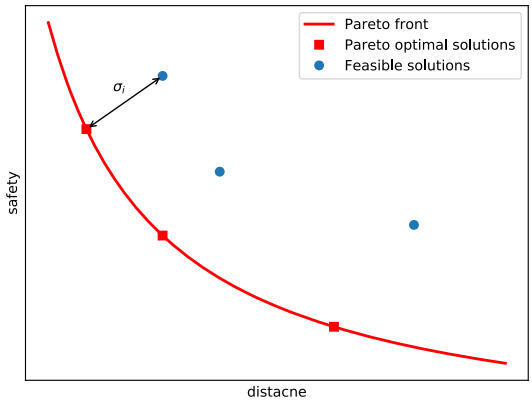


Fig. 9: The minimum distance to the Pareto optimal front

B. Results and Analysis

We carry out experiments on each network instance with 50 independent runs and calculate an average of obtained results. For an individual run, the source and destination nodes are randomly selected from the network.

Fig. 10 gives all solutions obtained when those algorithms meet the stopping condition. The figure shows that CDNSGA-II gets better accuracy in converging to the exact Pareto optimal front.

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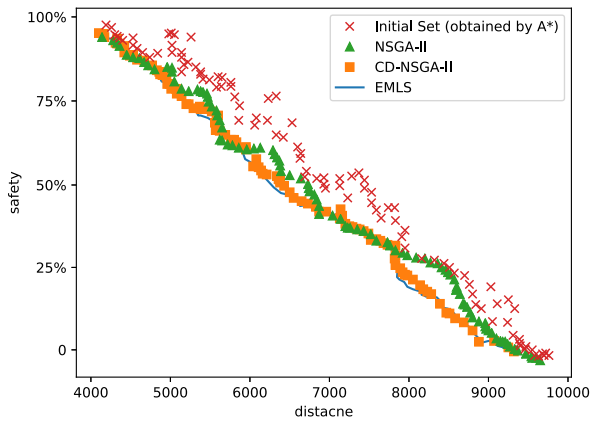


Fig. 10: Comparison of solutions

Fig. 11 shows planning paths for a UAV. There are two extreme paths (the shortest one and the safest one) and one trade-off path according to predefined weight ($\langle 0.4, 0.6 \rangle$).



Fig. 11: Planning paths for UAV

Table I shows the comparison of the solution quality. Comparing with NSGA-II, the proposed CDNSGA-II can obtain up to 80% Pareto optimal solutions even in a large-scale area.

TABLE I: Comparison of solution quality

Area	Optimal Solution Ratio (δ) / Average Error (ρ) $\times 10^{-2}$		
	EMLS	NSGA-II	CDNSGA-II
$AREA_1$	100% / 0	68.5%($\pm$2.5%) / 3.83	87.5%($\pm$1.4%) / 1.34
$AREA_2$	100% / 0	65.2%($\pm$2.3%) / 3.92	83.4%($\pm$1.9%) / 1.62
$AREA_3$	100% / 0	62.8%($\pm$2.5%) / 4.66	81.6%($\pm$1.5%) / 2.11

VI. Conclusion

This paper is concerned with the path planning problem for UAV under urban environment. An CDNSGA-II approach concerning two objectives, namely distance and safety, has been proposed. Meanwhile our approach uses an efficient data structure based on octrees. In particular, a SIM is established to indicate the main obstacles in the geography map. The results of synthetic and realistic experiments show that an effective path, which meets the safety requirement and meanwhile has a short distance, can be always suggested by the proposed framework under different cases.

Acknowledgment

This work was supported in part by the National Key Research and Development Program of China (No. 2017YFB1001900), the National Natural Science Foundation of China (No. 61876151, 61751208).

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